Machine Learning Approaches for Financial Transaction Fraud Detection

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Abstract. Financial fraud in electronic transaction systems presents a significant challenge due to the increasing volume of digital payments and the complexity of fraudulent behavior. This study investigates the application of machine learning techniques to detect fraudulent financial transactions using a large dataset of approximately 5 million records obtained from Kaggle.
The analysis follows a structured data science pipeline that includes data preparation, exploratory data analysis, feature analysis, model development, and evaluation. Key techniques include classification modeling using Logistic Regression and Random Forest, along with evaluation metrics such as accuracy, precision, recall, and F1-score.
The results indicate that model performance is strongly influenced by the skewed class distribution within the dataset. Although the Random Forest achieved high overall accuracy, it did not detect fraudulent transactions. The Logistic Regression model demonstrated better detection capability, achieving moderate recall but low precision, highlighting a tradeoff between identifying fraud and minimizing false positives.
Overall, the findings suggest that while machine learning models can support fraud detection efforts, additional techniques such as feature engineering, resampling methods, and model optimization are necessary to improve predictive performance in highly imbalanced financial datasets.

Keywords: financial fraud detection · anomaly detection · machine learning · financial transaction analysis

1 Introduction

The domain selected for this project is financial systems, which focuses on fraud detection in electronic financial transactions. As digital payment systems continue to expand globally, electronic transactions such as card payments, automated clearing house (ACH) transfers, wire transfers, and mobile payment systems have become increasingly common. Although these technologies provide convenience and efficiency, they have also created new opportunities for financial fraud [1, 3, 4, 6].

Investigating financial fraud often requires identifying patterns of abnormal transaction behavior across different payment systems. Law enforcement investigators frequently examine factors such as transaction location, transaction timing, transaction amounts, and unusual spending patterns to determine whether

fraudulent activity has occurred. According to a financial crime investigator with professional experience in fraud analysis (personal communication), abnormal transaction patterns and inconsistencies in financial activity are often key indicators used to initiate fraud investigations. These investigative practices highlight the importance of analyzing large transaction datasets to detect suspicious activity [5, 7].

Data for this project will be obtained from publicly available datasets hosted on Kaggle. The selected dataset contains anonymized financial transaction records in several types of electronic payments, including card transactions, ACH transfers, Unified Payments Interface (UPI) payments, and wire transfers. Additional research and background information will be gathered from academic sources such as Google Scholar and peer-reviewed articles on financial fraud detection and machine learning.

The data problem being investigated is the identification of fraudulent transactions within electronic payment systems. Fraudulent activity may differ from legitimate transactions based on characteristics such as payment method, transaction amount, transaction timing, and behavioral patterns. Detecting these patterns using machine learning techniques, including classification models and anomaly detection approaches, can help financial institutions identify suspicious transactions more quickly and reduce financial losses [2]. One challenge in fraud detection is that fraudulent transactions typically make up only a small portion of total transaction data [3]. Because financial fraud affects millions of individuals and organizations each year, improving fraud detection methods is an important and relevant problem in modern financial systems [1].

1.1 Goals of this Research

The goals of this project are to identify patterns associated with fraudulent financial transactions, evaluate the effectiveness of machine learning models for fraud detection, and examine whether fraud behavior differs between types of electronic payment. Each goal will be addressed through data analysis, model development, and comparison of transaction characteristics within the dataset.

1. **Identify patterns of fraudulent activity within electronic financial transactions** Through exploratory analysis of the dataset, this study will examine transaction characteristics such as payment type, transaction amount, and transaction frequency to identify patterns associated with fraudulent activity.
2. **Evaluate whether machine learning models can effectively detect fraudulent transactions** This goal will be addressed by training and evaluating machine learning classification models in the transaction dataset and comparing their performance using evaluation metrics such as accuracy, precision, recall, and F1-score [2].
3. **Examine how different electronic payment methods may influence fraud patterns** By analyzing fraud rates across different payment types such as ACH, card, UPI, and wire transfers, the study will evaluate whether

certain payment methods are more frequently associated with fraudulent activity.

2 Project Methodology

The implementation of this project will follow several phases in order to analyze financial transaction data and detect fraudulent activity.

Phase 1: Data Collection

The first phase will involve obtaining the Kaggle financial transaction dataset. This dataset contains anonymized records for several types of electronic payment, including card payments, automated clearing house (ACH) transfers, unified payment interface (UPI) payments, and wire transfers.

Phase 2: Data Preparation and Cleaning

The dataset will be examined for missing values, inconsistencies, and formatting issues. Any necessary cleaning will be performed to ensure that the data are suitable for analysis, as proper data preparation is critical to reliable machine learning performance [1].

Phase 3: Exploratory Data Analysis

Exploratory analysis will be performed to better understand the dataset and identify patterns within the transaction data. This may include examining transaction distributions, payment types, and differences between fraudulent and legitimate transactions.

Phase 4: Feature Analysis

Important transaction characteristics such as payment type, transaction amount, and transaction timing will be analyzed to determine which variables can help distinguish fraudulent transactions from legitimate ones.

Phase 5: Model Development

Machine learning models will be applied to classify transactions as fraudulent or legitimate, leveraging both supervised classification techniques and anomaly detection approaches to evaluate their effectiveness in fraud detection [5, 7, 2, 8].

Phase 6: Model Evaluation

The performance of the models will be evaluated using metrics such as accuracy, precision, recall, and F1-score. These metrics are commonly used to assess classification performance in fraud detection models [3].

Phase 7: Results Interpretation

The final phase will involve interpreting the results of the analysis and discussing how the findings can contribute to improved fraud detection in financial transaction systems.

3 Phase 1 Detailed Description: Data Collection

The data collection phase involves obtaining a large dataset of financial transaction records from Kaggle. The selected dataset contains approximately 5 million synthetically generated transactions and is provided in CSV format. It includes 18 attributes describing transaction details, behavioral features, metadata, and fraud indicators. Due to the size of the dataset, it was downloaded and stored on an external terabyte drive for efficient storage and processing.

This dataset was selected because it simulates realistic financial transaction behavior in multiple payment types, including card payments, ACH transfers, wire transfers, and digital payment systems. The inclusion of behavioral features and fraud labels makes it suitable for machine learning applications aimed at identifying fraudulent activity. Although the dataset is synthetically generated, it reflects common fraud patterns and provides a solid foundation for analysis.

4 Phase 2 Detailed Description: Data Preparation and Cleaning

The data preparation and cleaning phase focused on ensuring data quality, consistency, and usability prior to analysis and modeling. The dataset was examined for missing values, duplicate records, and inconsistencies in formatting to ensure accuracy and reliability.

The data cleaning workflow is illustrated in Figure 1.

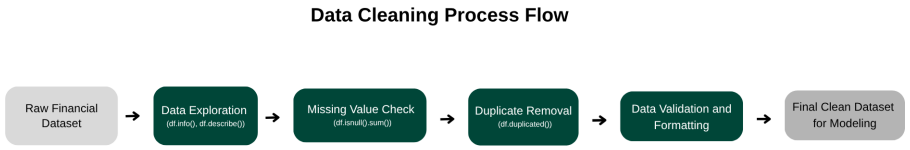

Fig. 1. Data cleaning process for the financial fraud dataset, including data exploration, validation, and preprocessing steps.

4.1 Data Curation Process

The data curation process focused on preparing the dataset for accurate analysis and modeling by ensuring data quality, consistency, and usability. The dataset was first explored to understand its structure, including the number of records, attributes, and data types.

During this process, checks were performed to identify missing values, duplicate records, and inconsistent formatting. Duplicate records were evaluated and removed where necessary to prevent bias in the analysis. Data types were also reviewed to ensure that numerical and categorical variables were properly formatted.

Although the dataset was largely clean, this step was essential to validate data integrity and confirm that the dataset was suitable for fraud detection modeling.

Data curation is important because raw datasets often contain missing values, duplicates, and inconsistencies that can negatively impact analysis. By cleaning and organizing the data, the dataset becomes more accurate and reliable, which is essential to building effective fraud detection models.

4.2 Tools and Techniques

The data cleaning and preprocessing steps were performed using Python in a Jupyter Notebook environment. The primary libraries used were Pandas and NumPy, which provided efficient tools for data manipulation and analysis.

Several techniques were applied during preprocessing, including:

- Use of `df.info()` and `df.describe()` to examine the structure of the dataset and summary statistics
- Identification of missing values using `df.isnull().sum()`
- Detection of duplicate records using `df.duplicated()`
- Validation of data types to ensure consistency across variables

These tools and techniques allowed for a structured and systematic approach to prepare the dataset for analysis.

4.3 Handling Missing Values

Missing values were assessed using `df.isnull().sum()`, which indicated that the dataset contained little to no missing data. As a result, no records or attributes required removal or imputation.

Although missing values were minimal, several strategies were considered in case missing data had been present. These strategies include removing rows or columns with excessive missing values, imputing missing values using statistical methods such as mean, median, or mode, and applying domain-based interpretation to determine appropriate replacements.

The decision to retain all data without modification was based on the completeness of the dataset and the importance of preserving the integrity of the data for the analysis of fraud detection.

4.4 Final Dataset Size

After completing the data cleaning process, the final dataset consisted of approximately 5,000,000 records and 18 attributes.

Since minimal cleaning was required, the majority of the original dataset was retained. This ensured that sufficient data was available for analysis and model development while maintaining the integrity of the dataset.

The dataset is highly imbalanced, with significantly fewer fraudulent transactions than non-fraudulent ones, which is an important consideration for model performance.

4.5 Definitions of Essential Attributes

Several key attributes were identified as essential for the fraud detection analysis.

- **Time**: Represents the time elapsed between each transaction and the first transaction in the dataset.
- **Amount**: Indicates the monetary value of each transaction.
- **V1–V18**: Transformed variables representing transaction features and behavioral patterns.
- **Class**: The target variable indicating whether a transaction is fraudulent (1) or non-fraudulent (0).

These attributes are critical for identifying patterns and distinguishing between legitimate and fraudulent transactions.

4.6 Variable Identification

The dependent variable in this analysis is the **Class** attribute, which represents whether a transaction is fraudulent or not.

The independent variables include all other features in the dataset, such as **Time**, **Amount**, and transformed variables **V1–V18**. These variables are used to identify patterns and relationships that contribute to detecting fraudulent activity.

By analyzing the relationship between the independent and dependent variables, machine learning models can be trained to classify transactions accurately.

5 Phase 3 Detailed Description: Exploratory Data Analysis

The exploratory data analysis (EDA) phase focused on understanding the structure, distribution, and key characteristics of the dataset prior to modeling. This phase included data inspection, cleaning, descriptive statistical analysis, visualization, and outlier detection to identify patterns and potential anomalies within the transaction data.

The EDA workflow is illustrated in Figure 2.

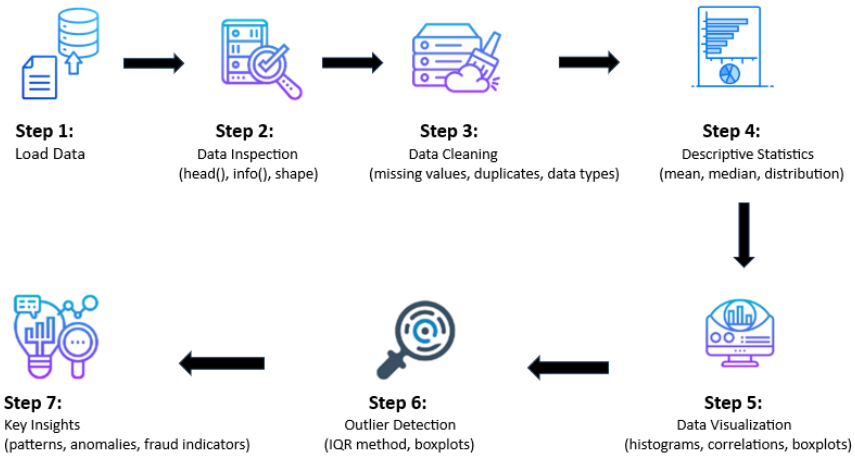

Fig. 2. Exploratory Data Analysis (EDA) workflow applied to the financial fraud dataset, including data inspection, cleaning, statistical analysis, visualization, and outlier detection to identify patterns, anomalies, and potential fraud indicators.

5.1 Definition and Importance

Exploratory Data Analysis (EDA) is the process of examining and understanding a dataset through statistical summaries and visualizations prior to model development. It is used to identify patterns, detect anomalies, and assess data quality.

EDA is essential because it ensures that the dataset is suitable for analysis and helps identify potential challenges such as class imbalance, outliers, and skewed distributions that may impact model performance.

5.2 EDA Techniques

The following exploratory data analysis techniques were applied:

- Descriptive statistics using `df.describe()`
- Data structure analysis using `df.info()`
- Missing value analysis using `df.isnull().sum()`
- Duplicate detection using `df.duplicated()`
- Univariate analysis of key variables such as the transaction amount
- Class distribution analysis of fraudulent vs non-fraudulent transactions
- Outlier detection through statistical observation

5.3 Techniques and Results

Exploratory data analysis techniques were applied to better understand the characteristics of the dataset and identify patterns related to fraudulent activity. Descriptive statistics revealed that the dataset contains approximately 5 million transactions across 18 attributes, including both numerical and transformed variables.

Univariate analysis of key variables, such as transaction amount, showed a highly skewed distribution. Most transactions were low in value, while a small number of transactions had significantly higher amounts. These extreme values may indicate unusual or potentially fraudulent behavior.

Analysis of the target variable (Class) revealed a significant imbalanced data structure, where fraudulent transactions represent a small fraction of the total dataset. This is a common challenge in fraud detection and can affect model performance if not addressed adequately [3].

Duplicate detection confirmed that only a minimal number of duplicate records were present and removed to maintain data integrity. Additionally, missing value analysis showed that the dataset contained little or no missing data, allowing the majority of the data to be retained without imputation.

The outlier analysis suggested that unusually large transaction amounts and abnormal patterns in transformed variables may be associated with fraudulent activity. These findings indicate that certain features, such as the amount of transaction and the behavioral variables, are important in the identification of fraud.

These techniques provided key insights into the structure and patterns of data relevant to fraud detection, highlighting key characteristics that will influence the development of the model.

To further support the statistical findings, visualizations were generated to examine the distribution and behavior of key variables.

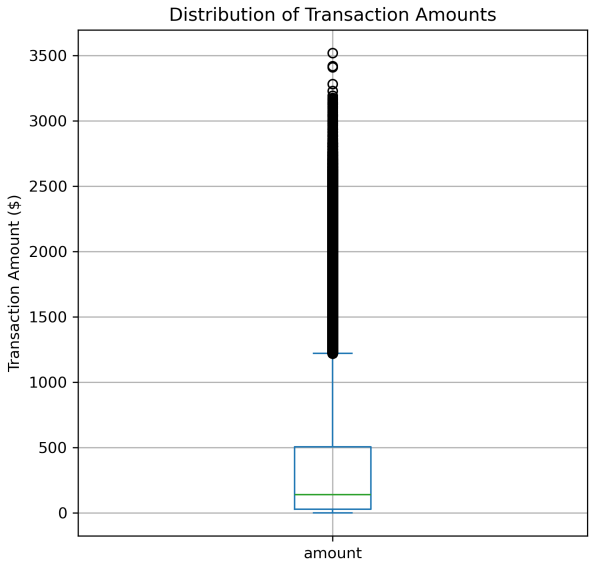

Fig. 3. Distribution of transaction amounts showing right-skewed pattern with numerous high-value outliers.

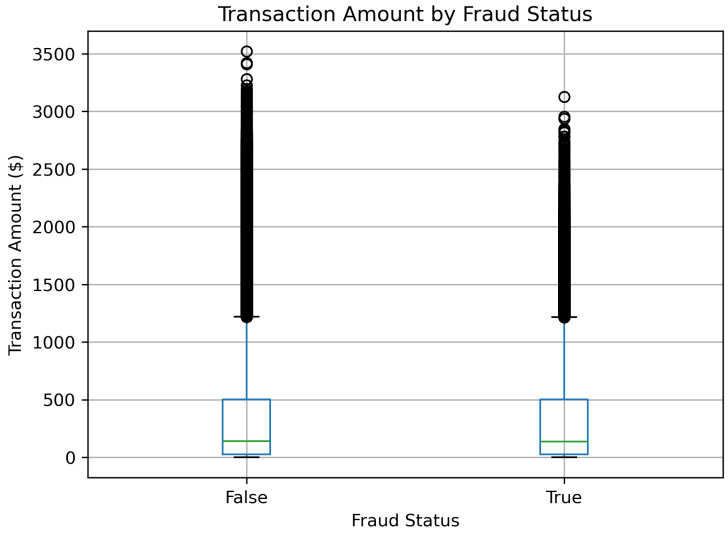

Fig. 4. Transaction amounts grouped by fraud status. The distributions show substantial overlap, indicating that transaction amount alone is not a strong predictor of fraud and that additional features are required for accurate detection.

The comparative analysis shows that both fraudulent and non-fraudulent transactions exhibit similar distributions with significant overlap. Although some

variability is present, the results suggest that transaction amount alone is insufficient to distinguish fraudulent activity.

Additional analysis of other variables, such as the velocity score, indicated more uniform distributions with limited variability between fraud classes. This further emphasizes the need to evaluate multiple features in combination when developing predictive models.

In general, the EDA phase provided critical insights into data distribution, variability, and feature behavior, which will inform subsequent modeling and feature selection steps in the project.

5.4 Key Learning

The exploratory data analysis phase provided several important insights into the dataset. One of the most significant findings was the presence of an imbalanced data structure, which highlights the need for careful model evaluation and the potential use of resampling techniques in later stages.

Identifying skewed distributions and outliers emphasized the importance of data transformation and feature scaling to improve model performance. Additionally, the analysis confirmed that the dataset is largely complete and suitable for machine learning applications.

This phase reinforced the need to identify data challenges prior to model development. By identifying patterns, anomalies, and potential challenges early in the process, more informed decisions can be made during feature selection and model design.

In general, the EDA phase improved the understanding of the dataset and provided a solid foundation for building effective fraud detection models.

6 Limitations

Several limitations were identified in this study. First, the dataset used for analysis was synthetically generated and anonyomized, which may not fully capture the complexity and variability of real-world financial transaction data. As a result, the findings may not be generalized directly to operational fraud detection systems.

Second, the dataset exhibited significant class imbalance, with fraudulent transactions representing only a small fraction of the total data. This imbalance negatively impacted model performance, particularly in precision and recall, and limited the effectiveness of classification models.

Additionally, the models used in this study were relatively basic and did not incorporate advanced techniques such as hyperparameter tuning, ensemble optimization, or deep learning approaches. These methods may improve performance in future work.

Finally, feature availability was limited to the variables provided in the dataset. In real-world scenarios, additional contextual and behavioral features may enhance the ability to detect fraudulent transactions more accurately.

7 Phase 4 Detailed Description: Predictive Analysis and Model Building

7.1 Predictive Pipeline

The predictive pipeline was designed to detect fraudulent financial transactions using a structured machine learning workflow. The pipeline begins with data preprocessing, where missing values were handled, categorical variables were encoded, and numerical features were scaled where appropriate.

Following preprocessing, feature selection was performed to retain relevant variables such as transaction amount, transaction type, location, device used, and behavioral indicators (e.g., velocity score and geo anomaly score).

The processed data was then split into training and testing subsets. Machine learning models were trained using the training data and evaluated on unseen test data. The model predictions were compared with the actual labels to assess performance.

The pipeline consists of the following steps:

1. Data cleaning and preprocessing
2. Feature engineering and selection
3. Data splitting (training and testing)
4. Model training
5. Model evaluation
6. Performance comparison

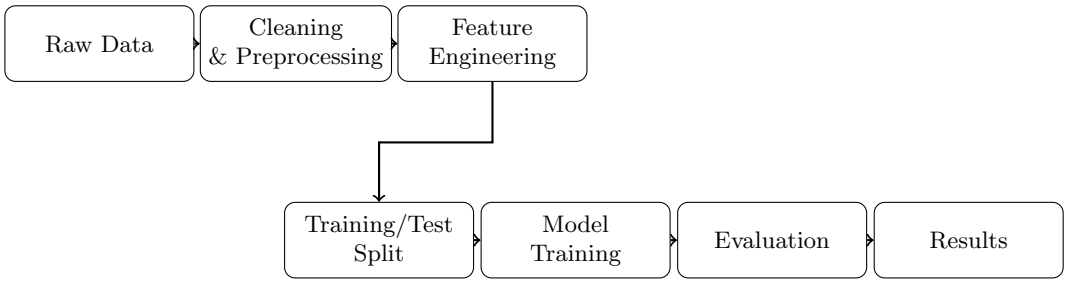

Fig. 5. Predictive modeling pipeline for fraud detection, illustrating data preprocessing, feature engineering, model training, and evaluation stages.

7.2 Machine Learning Algorithms

Several machine learning algorithms were selected to analyze the fraud detection dataset and compare their performance. These models were chosen because they are commonly used for classification problems and provide a useful comparison between simple and more advanced approaches.

Logistic Regression was used as a baseline classification model. It estimates the probability that a transaction is fraudulent based on the input features. This model is simple and efficient and provides a useful benchmark for comparison.

Decision Tree was applied to capture non-linear relationships within the data. It works by splitting the dataset into branches based on feature values, making it easy to interpret and useful for identifying decision rules.

Random Forest was used as an ensemble learning method. It combines multiple decision trees to improve prediction accuracy and reduce overfitting. This model is particularly effective in fraud detection tasks due to its ability to handle complex interactions between variables.

These algorithms provided a range of modeling approaches, allowing performance to be compared between interpretable and ensemble-based methods.

7.3 Training and Testing Process

The dataset was divided into training and testing subsets using an 80/20 split. The training set was used to train machine learning models, while the testing set was reserved for evaluating model performance on unseen data.

Due to the presence of class imbalance, where fraudulent transactions are significantly fewer than non-fraudulent transactions, techniques such as class weighting and resampling were considered. These approaches help prevent the models from becoming biased toward the majority class.

This process ensures that the models are evaluated for their ability to generalize to new data rather than simply memorizing patterns from the training dataset.

7.4 Implementation and Evaluation

The models were implemented using Python with libraries including pandas, NumPy, and scikit-learn. Each model was trained using the training dataset and evaluated using the testing dataset.

The performance of the model was assessed using the following evaluation metrics.

- **Accuracy:** Measures the overall correctness of the model
- **Precision:** Measures how many predicted fraud cases were actually fraudulent
- **Recall:** Measures the ability to identify actual fraudulent transactions
- **F1-Score:** Provides a balance between precision and recall

In fraud detection, recall and precision are particularly important because failing to detect fraudulent transactions can result in significant financial loss.

The source code for this project is available on GitHub: https://github.com/14dstcyr/fraud-detection-project.

7.5 Results

The performance of the Logistic Regression model is summarized in Table 1, while Table 2 presents a comparative analysis of the models.

Table 1. Logistic Regression Model Performance

Metric	Value
Precision	0.04
Recall	0.48
F1-Score	0.07
Accuracy	0.52

Table 2. Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.52	0.04	0.48	0.07
Random Forest	0.96	0.00	0.00	0.00

The Logistic Regression model was able to detect fraudulent transactions, achieving a recall of approximately 0.48. However, the precision was low (0.04), indicating a high number of false positives. This suggests that while the model is somewhat effective in identifying fraud cases, it incorrectly classifies many non-fraudulent transactions as fraudulent.

The Random Forest model achieved a high overall accuracy of approximately 0.96, but failed to detect any fraudulent transactions. This highlights a key limitation of using accuracy as the sole evaluation metric in imbalanced datasets, as the model can achieve high accuracy simply by predicting the majority class.

Threshold tuning was applied to the Logistic Regression model to explore the tradeoff between recall and precision. Lowering the classification threshold increased recall to 1.00, which means that all fraudulent transactions were identified. However, this resulted in a significant decrease in precision, as nearly all transactions were classified as fraudulent. This demonstrates the tradeoff between detecting fraud and minimizing false positives, which is a well-documented challenge in fraud detection systems [3, 1].

8 Interpretation of Results and Conclusions

8.1 Charts and Visualizations

Several types of charts are appropriate to highlight the results of a fraud detection model, particularly those that evaluate classification performance and model effectiveness. Common visualization techniques include confusion matrices, ROC curves, precision-recall curves, and model comparison charts.

In this analysis, the main visualizations used were the confusion matrix, the ROC curve, and a model comparison bar chart. These charts were selected because they provide clear insight into the accuracy, error distribution, and overall performance of the model across different algorithms.

8.2 Results Explanation

The confusion matrix (Figure 6) provides a detailed breakdown of the model predictions, true negatives, false positives, and false negatives. This was especially important in evaluating how the model identified fraudulent transactions, as false negatives represent missed fraud cases.

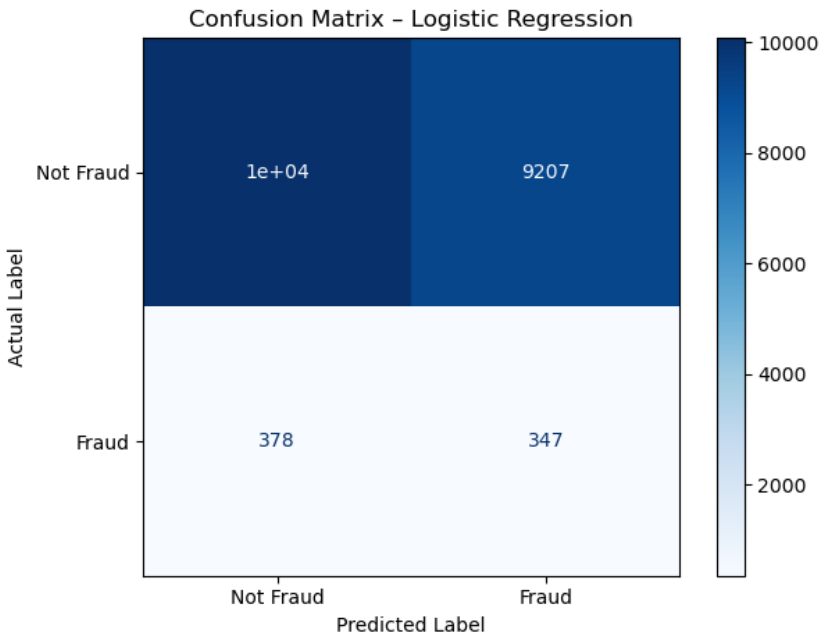

Fig. 6. Confusion matrix for the Logistic Regression model, showing true positives, true negatives, false positives, and false negatives.

The ROC curve (Figure 7) demonstrated the trade-off between the true positive rate and false positive rate, showing the model's ability to distinguish between fraudulent and non-fraudulent transactions. A higher area under the curve (AUC) indicated better model performance.

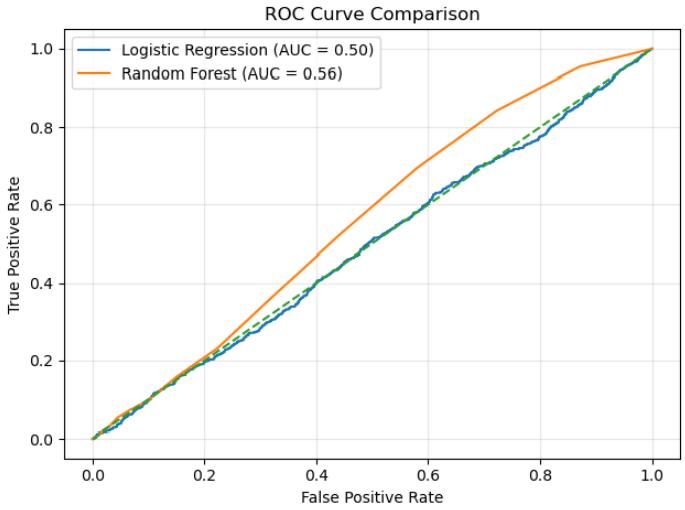

Fig. 7. ROC curve illustrating the trade-off between true positive rate and false positive rate for the Logistic Regression model.

Additionally, the model comparison chart (Figure 8) showed differences in performance metrics such as accuracy, precision, recall, and F1-score between the algorithms tested. This allowed for direct comparison and identification of the most effective model.

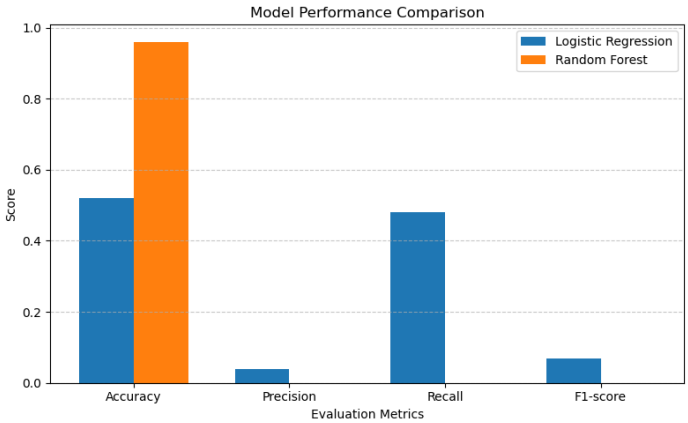

Fig. 8. Comparison of model performance metrics including accuracy, precision, recall, and F1-score between different models.

The feature importance analysis (Figure 9) highlights the variables that most influenced model predictions.

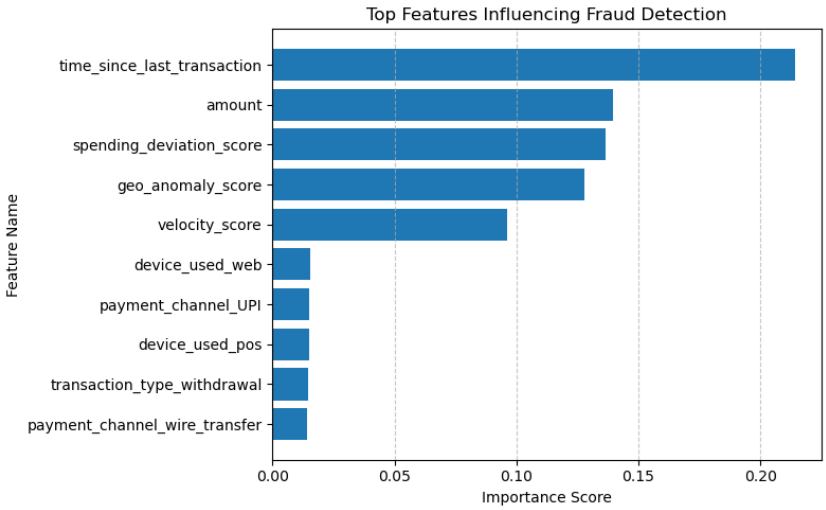

Fig. 9. Top features influencing fraud detection based on the Random Forest model.

Temporal and behavioral features, such as the time since the last transaction, the number of transactions, and the spending deviation score, were among the most influential variables. However, despite these insights, the Random Forest model did not effectively detect fraudulent transactions, suggesting that these features alone were insufficient for accurate classification.

8.3 Data Inference

From the analysis, it was observed that the dataset exhibited a skewed class distribution, with significantly fewer fraudulent transactions compared to the legitimate ones. This influenced the performance of the model, particularly in recall and precision.

The models were able to achieve high accuracy; however, accuracy alone was not a sufficient measure due to the minority fraud class. Models that demonstrated higher recall were more effective in identifying fraudulent transactions, which is critical in this domain.

8.4 Statistical Conclusions

Based on the evaluation metrics, neither model demonstrated strong predictive performance for fraud detection. Although the Random Forest model achieved high overall accuracy, it did not identify any fraudulent transactions, as indicated by a recall of 0.00 for the fraud class. This highlights the limitations of relying solely on accuracy in imbalanced datasets.

Logistic Regression showed better fraud detection capacity, achieving a recall of approximately 0.48; however, precision was very low (0.04), indicating a high number of false positives. This trade-off suggests that while the model can identify some fraudulent activity, it does so at the cost of incorrectly flagging many legitimate transactions.

Collectively, the results indicate limited classification performance, suggesting that additional techniques, such as feature engineering, resampling methods, or more advanced modeling approaches, may be required to improve fraud detection performance.

8.5 General Observations

In general, the analysis demonstrates that machine learning models can support fraud detection; however, performance remains limited in highly imbalanced datasets. Proper preprocessing and the selection of appropriate evaluation metrics were critical to achieving meaningful results.

Additionally, the results highlight the importance of aligning model performance with business objectives. In the context of fraud detection, prioritizing recall is essential to reduce the risk of undetected fraudulent activity.

These findings support the practical application of predictive modeling techniques in real-world financial systems, where accurate and timely fraud detection is critical.

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